Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Java/Python/C++ applications**, relational data systems, and cross-platform software; experienced across requirements, design updates, testing, installation, deployment, user support, root-cause analysis, corrective action, documentation, and Agile collaboration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Coordinated with users to **translate requirements into application updates**, developing Python services and relational structures for 500+ users while optimizing indexed queries to reduce latency from 1.2 seconds to 350 milliseconds.
- Performed **root-cause analysis and corrective action** across 5,000+ monthly workflows, documenting malformed inputs and failed integrations while implementing validation, reconciliation, error handling, and CI/CD releases that achieved 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented **design changes and application features** across React interfaces and Flask services for uploads, processing, history, and reporting, supporting 150+ users and 10,000+ daily requests through maintainable MVC architecture.
- Assisted with **testing, deployment, and user support**, debugging file-processing defects, incorporating code-review feedback, and shipping GitHub Actions-backed updates that reduced turnaround time 60% while maintaining reliable production performance.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed **Python simulation and evaluation software** processing 100,000+ autonomous-system trajectories, integrating map geometry and temporal signals while defining reproducible test scenarios that improved long-horizon planning performance 18%.
- Analyzed **system requirements, benchmark results, and failure patterns** across three architectures, preparing technical documentation and recommending software changes that improved simulated closed-loop stability 15% through quantitative performance feedback.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Maintained and refactored **object-oriented API and data components** using Singleton and Factory patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and maintainability.
- Managed **application authoring, installation, testing, and deployment workflows** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while validating 20+ tests across Windows, Linux, and macOS releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed and integrated **React, Node.js, and Azure SQL applications** across eight services for authentication, records, billing, permissions, and notifications, synchronizing 2M+ records through REST APIs and real-time WebSockets.
- Collaborated on **design, code reviews, testing, and deployment** using Jest, Selenium, Sentry, Docker, and Jenkins CI/CD, debugging defects and cutting release time 50% while improving program consistency and reliability.

Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Applications: Spring Boot, FastAPI, Flask, Node.js, React.js, MVC, REST APIs, WebSockets, microservices
Data: SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, relational design, indexing, validation, reconciliation
Lifecycle: Requirements, application updates, testing, installation, deployment, user support, Git, Jenkins, GitHub Actions, CI/CD
Engineering: OOP, design patterns, troubleshooting, root-cause analysis, corrective action, documentation, Linux, Docker, Agile/Scrum

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Enhanced a **cross-platform database application** across Java and TypeScript modules, supporting 10+ SQL/NoSQL systems with schema exploration, query generation, data management, Docker, Maven, and reusable desktop development workflows.

CPU Scheduling Simulator Live | *C++, React.js, Data Structures, Algorithms*
- Implemented and tested **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Deployed a **containerized Python application** to AWS ECS/Fargate using CodePipeline, automated validation, and versioned CI/CD, achieving 87% accuracy and reliable, repeatable builds, installation, testing, and releases.